

Literature Status Packet: the uncountable Kottman question

FA/Banach run `fa.banach.001`

2026-06-14

Status

Hajek–Kania–Russo arXiv:1803.11501 asks whether the unit sphere of every nonseparable Banach space contains an uncountable $(1+)$ -separated subset. Koszmider arXiv:2104.05335 later gives a negative answer by constructing a nonseparable Banach space whose uncountable sphere subsets always contain a large subset with all mutual distances below 1.

This packet is classified as `literature_already_answered`. It is not an original proof from this run.

Original Problem Source

Original source:

- Petr Hajek, Tomasz Kania, and Tommaso Russo, *Separated sets and Auerbach systems in Banach spaces*, arXiv:1803.11501; Trans. Amer. Math. Soc. 373 (2020), no. 10, 6961–6998.

In the introduction, the paper formulates the nonseparable analogue of Kottman’s theorem: must every nonseparable Banach space have an uncountable $(1+)$ -separated subset of its unit sphere? The authors state there that the answer remains unknown, then prove many positive results for special classes such as reflexive, WLD under additional density assumptions, RNP, and strictly convex spaces.

Supporting Answer Source

Supporting source:

- Piotr Koszmider, *Banach spaces in which large subsets of spheres concentrate*, arXiv:2104.05335.

Koszmider’s abstract says the paper solves the central nonseparable Kottman problem negatively. In the introduction, the paper describes the question whether every nonseparable Banach-space sphere has an uncountable $(1+)$ -separated subset, cites Hajek–Kania–Russo as highlighting this central question, and then states the negative theorem.

Idea Of The Answer

The construction uses a Johnson–Lindenstrauss space generated by an $\mathbb{R}$ -embeddable almost disjoint family. In the original supremum norm, the special combinatorics of the almost disjoint family force large subsets of the sphere to contain equally large subsets whose pairwise distances are at

most $1 + \varepsilon$. A strictly convex renorming obtained through a bounded injective map into a separable space then sharpens this into concentration below 1 on uncountable subsets of regular cardinality.

Thus the resulting nonseparable space cannot contain an uncountable $(1+)$ -separated subset of its unit sphere. The result is stronger than the negation of the original question: the same space has no uncountable equilateral set and no uncountable Auerbach system.

Verification Notes

- Same-paper check: arXiv:1803.11501 records the general question as open and gives only class-specific positive results.
- Separate-source check: arXiv:2104.05335 is a later paper and cites the Hajek–Kania–Russo source in its discussion of the central question.
- Logical implication: an uncountable $(1+)$ -separated subset has every two distinct points at distance > 1 . Koszmider’s space has the opposite large-subset behavior: every uncountable subset of the sphere contains an uncountable subset with all pairwise distances < 1 .
- Therefore Koszmider’s theorem gives a negative answer to the exact problem extracted from arXiv:1803.11501.

Search Bounds

Before promotion, the run registry, solution index, attempt index, proof-gap index, active claims, and the arXiv:1803.11501 source text were checked for duplicate coverage. Web searches on 2026-06-14 for the exact question, **uncountable Kottman’s theorem**, **uncountable $(1+)$ -separated**, and the original paper title found arXiv:2104.05335 as the direct later answer.

Human Review Recommendation

Check the original introduction of arXiv:1803.11501 and the abstract, introduction, and Theorem 1 of arXiv:2104.05335. The review should focus on the source identification, not on re-proving Koszmider’s construction.